

AKHALTSIKHE, Georgia

WMO: 375140

Lat: 41.6506N

Lon: 43.0069E

Elev: 3262

StdP: 13.04

Time Zone: 4.00 (E04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.8	(c) 11.8	(d) 2.9	(e) 7.2	(f) 8.5	(g) 7.2	(h) 9.0	(i) 13.3	(j) 18.8	(k) 41.0	(l) 16.1	(m) 37.4	(n) 0.0	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 27.2	(c) 91.4	(d) 70.8	(e) 88.0	(f) 69.3	(g) 84.8	(h) 67.8	(i) 71.9	(j) 88.2	(k) 70.3	(l) 85.5	(m) 68.7	(n) 82.7	(o) 6.6	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.8	111.2	79.5	65.2	105.3	77.5	63.9	100.3	75.7	38.0	89.0	36.4	85.7	35.0	82.8	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.1	(b) 15.7	(c) 12.3	(e) 2.7	(f) 97.2	(g) 6.6	(h) 2.7	(i) -2.0	(j) 99.1	(k) -5.8	(l) 100.7	(m) -9.5	(n) 102.3	(o) -14.3	(p) 104.2		
(a) 19.1	(b) 15.7	(c) 12.3	(e) 2.2	(f) 74.5	(g) 6.5	(h) 1.6	(i) -2.4	(j) 75.7	(k) -6.2	(l) 76.6	(m) -9.8	(n) 77.5	(o) -14.5	(p) 78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	29.1	33.1	41.5	49.6	58.8	66.4	71.4	72.3	63.6	53.1	40.7	32.6
	DBStd	16.15	7.96	7.94	6.33	6.47	5.16	4.54	4.58	5.09	6.02	5.52	6.18	7.18
	HDD50	2343	648	474	271	87	4	0	0	1	35	282	541	
	HDD65	5652	1113	894	729	464	201	37	7	93	371	729	1006	
	CDD50	2744	0	0	7	73	276	493	663	691	409	129	3	0
	CDD65	577	0	0	0	1	8	79	205	233	51	0	0	0
	CDH74	5820	0	0	0	35	216	784	1802	2329	639	15	0	0
	CDH80	2088	0	0	0	4	32	201	670	1007	174	0	0	0
	WSAvg	2.2	1.1	1.8	3.2	2.7	2.3	3.0	3.0	3.0	2.2	1.3	1.2	1.2
	PrecAvg	23.5	1.2	1.1	1.5	2.2	3.2	3.1	2.5	2.0	1.8	1.9	1.6	1.2
(15) Precipitation	PrecMax	33.2	3.5	2.4	3.1	3.5	5.1	5.0	5.2	4.8	3.1	3.9	5.1	3.0
	PrecMin	17.2	0.0	0.2	0.5	1.0	1.7	0.7	0.2	0.7	0.5	0.3	0.0	0.2
	PrecStd	3.8	0.7	0.6	0.7	0.7	1.0	1.1	1.3	1.0	0.8	0.9	1.3	0.6
	PrecStd	3.8	0.7	0.6	0.7	0.7	1.0	1.1	1.3	1.0	0.8	0.9	1.3	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	49.6	57.1	67.8	78.2	83.5	88.9	94.1	96.3	89.9	76.0	63.1	54.7	
	0.4% MCWB	39.6	44.2	52.1	60.5	64.2	68.4	71.6	73.2	69.1	59.4	49.5	44.6	
	2% DB	45.2	52.5	62.3	72.6	79.1	85.0	90.5	92.7	84.7	72.0	58.7	49.1	
	2% MCWB	37.4	41.9	48.6	56.4	61.5	67.2	70.7	71.5	66.5	57.5	47.8	40.7	
	5% DB	42.0	48.5	58.4	68.6	75.4	81.7	86.9	89.2	80.8	68.6	54.9	45.6	
	5% MCWB	35.8	40.1	46.7	54.1	59.9	65.9	69.5	70.0	64.6	56.0	45.9	38.8	
	10% DB	39.0	44.7	54.8	64.3	71.5	78.2	83.5	85.7	77.2	64.9	51.2	42.3	
	10% MCWB	34.3	37.7	44.7	51.7	58.1	64.1	67.9	68.4	62.6	54.3	44.2	37.2	
	0.4% WB	40.8	44.9	53.3	62.7	66.1	70.2	73.3	75.4	71.0	60.7	51.4	45.9	
	0.4% MCDB	46.7	54.0	64.6	76.7	79.2	85.9	91.0	92.8	85.0	73.2	59.9	54.3	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	38.5	42.7	49.6	57.3	62.9	68.2	71.5	72.7	67.6	58.6	48.9	41.7	
	2% MCDB	43.6	51.3	59.9	70.2	76.3	82.7	88.0	89.7	82.5	69.4	56.0	47.2	
	5% WB	36.6	40.6	47.5	54.9	60.7	66.5	70.0	70.9	65.2	56.9	47.0	39.6	
	5% MCDB	40.8	47.5	57.0	67.1	73.4	79.7	85.1	86.4	78.7	66.4	53.0	44.0	
	10% WB	34.9	38.4	45.3	52.6	58.9	64.9	68.5	69.2	63.2	55.2	45.2	37.9	
	10% MCDB	38.4	44.0	53.5	62.7	69.7	76.5	81.6	83.6	75.1	62.9	50.5	41.9	
(35) Mean Daily Temperature Range	MDBR	16.9	19.4	22.5	24.9	23.9	24.0	23.9	27.2	26.3	23.3	21.3	17.6	
	5% DB	19.0	24.2	30.2	33.9	32.8	29.3	30.0	32.9	32.8	30.6	27.4	21.6	
	5% WB	12.2	14.7	17.6	17.9	15.5	12.7	12.2	13.7	15.1	16.1	16.3	13.8	
	MCDBR	16.4	22.5	27.4	31.2	29.5	27.3	27.7	31.1	31.4	27.3	24.4	19.2	
	MCWBR	10.7	13.7	16.3	16.8	14.5	12.2	11.7	13.4	14.9	14.2	14.8	12.4	
	MCWBR	10.7	13.7	16.3	16.8	14.5	12.2	11.7	13.4	14.9	14.2	14.8	12.4	
(40) Clear-Sky Solar Irradiance	taub	0.305	0.345	0.407	0.468	0.431	0.411	0.444	0.446	0.382	0.378	0.325	0.298	
	taud	2.200	2.079	1.986	1.950	2.149	2.274	2.203	2.192	2.358	2.288	2.361	2.289	
	Ebn at Noon	262	266	262	255	269	274	263	259	270	254	255	256	
	Edh at Noon	30	41	50	56	48	42	45	44	35	33	26	26	
	RadAvg	604	943	1212	1476	1663	1882	1904	1777	1439	964	651	504	
(45) All-Sky Solar Radiation	RadStd	64	108	134	142	157	129	161	134	128	96	86	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page